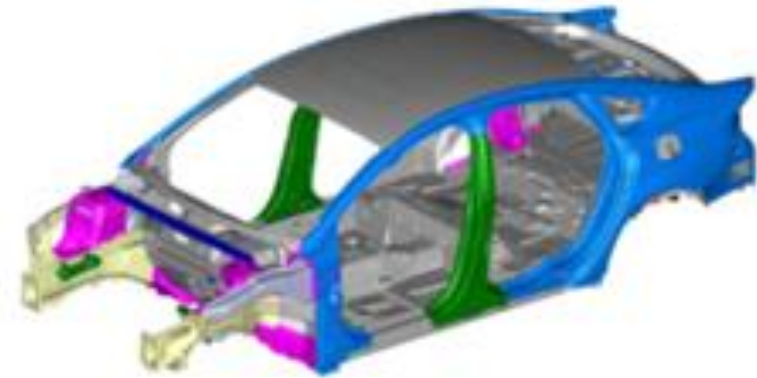


AUTOMOTIVE BIW PRODUCT DESIGN



BIW PRODUCT DESIGN COURSE

Week number

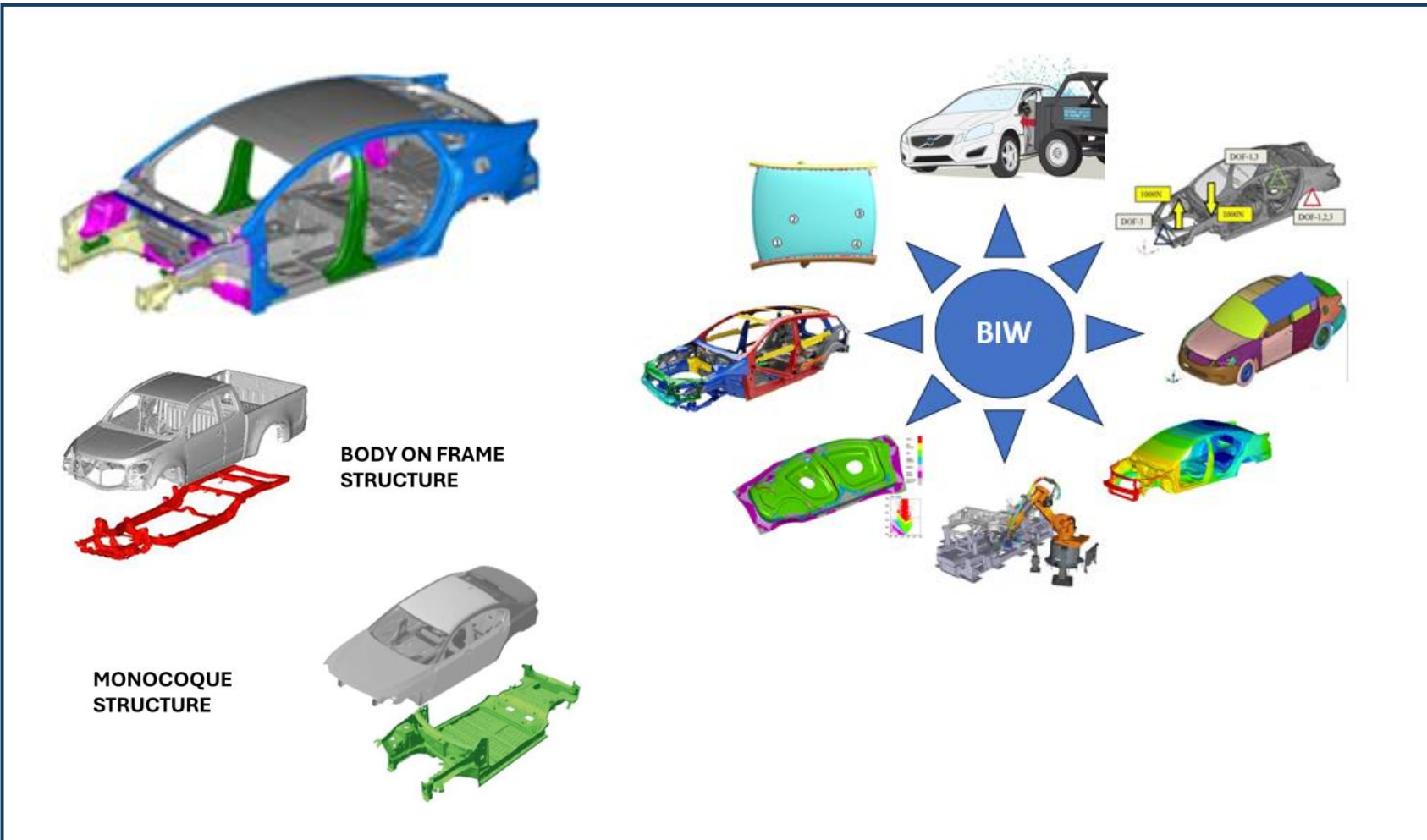
01

Day number

01

Topic name : BIW Introduction

Aim: To learn the basics of BIW design and functional requirements



BIW PRODUCT DESIGN COURSE

Week number

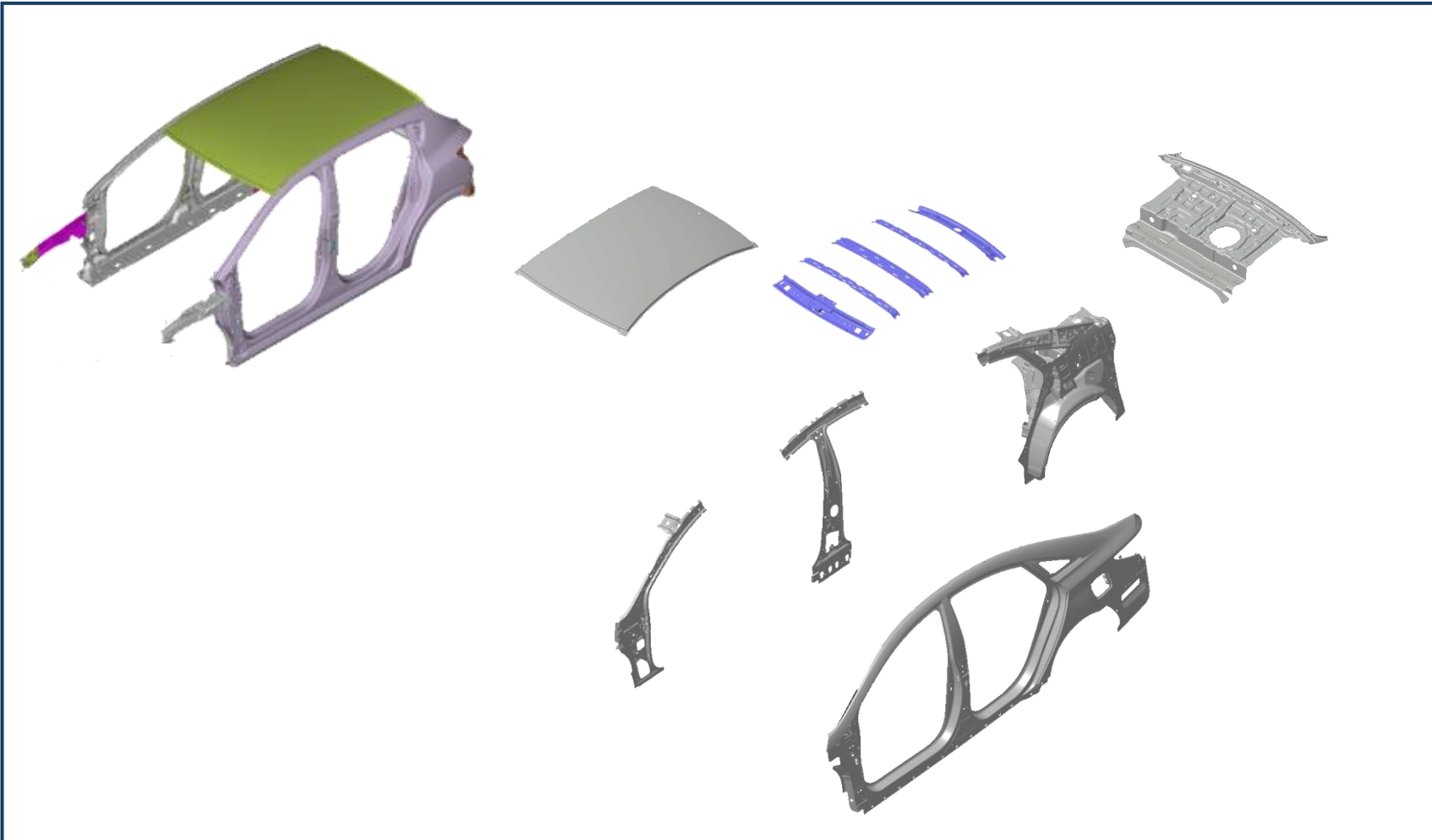
01

Day number

02

Topic name : BIW Nomenclature – Upper body

Aim: To learn the BIW upper body parts and functional requirements



BIW PRODUCT DESIGN COURSE

Week number

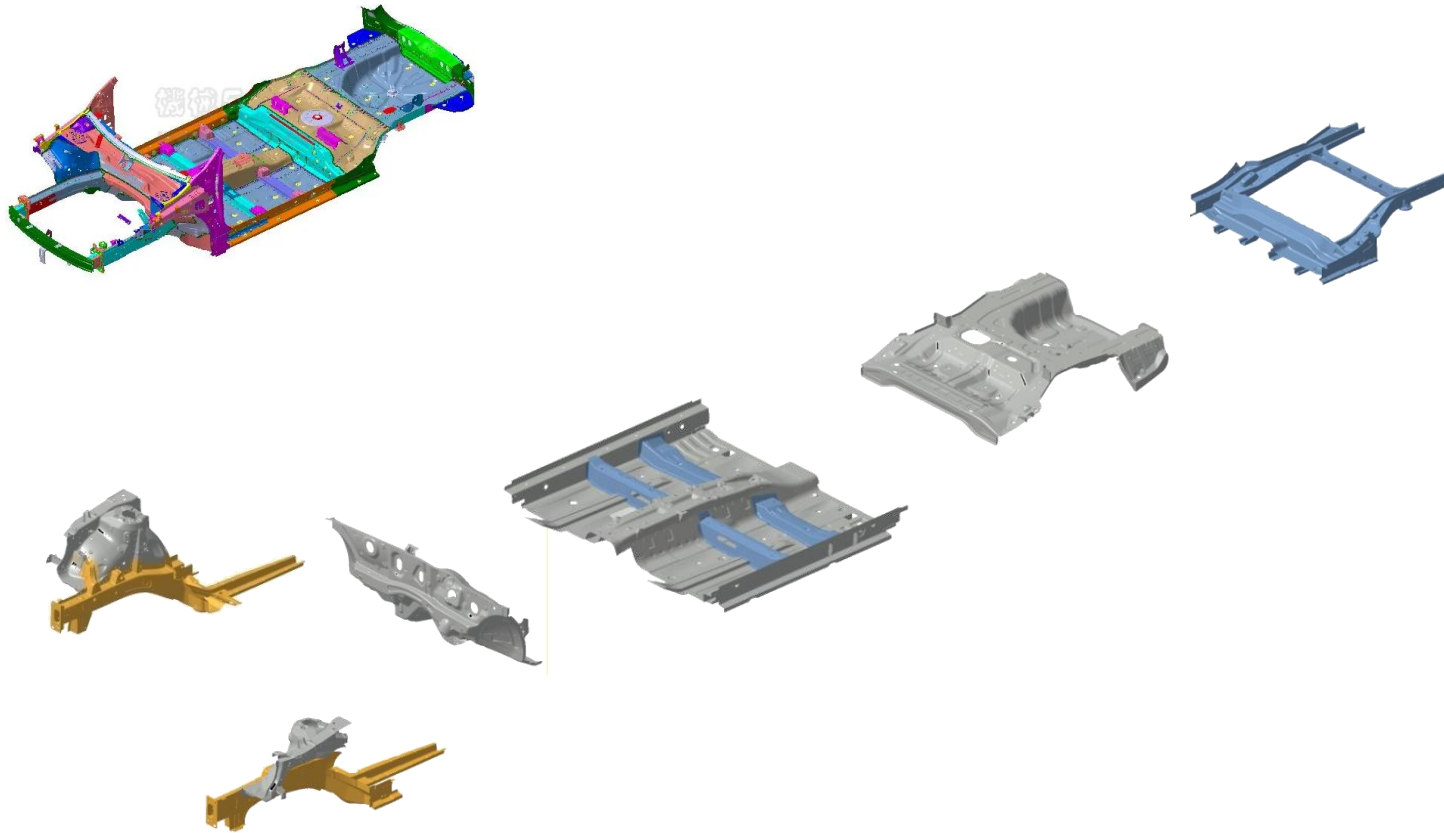
01

Day number

03

Topic name : BIW Nomenclature – Under body

Aim: To learn the BIW under body parts and functional requirements



BIW PRODUCT DESIGN COURSE

Week number

02

Day number

04

Topic name : BIW Nomenclature – Closures

Aim: To learn the BIW closure parts and functional requirements



BIW PRODUCT DESIGN COURSE

Week number

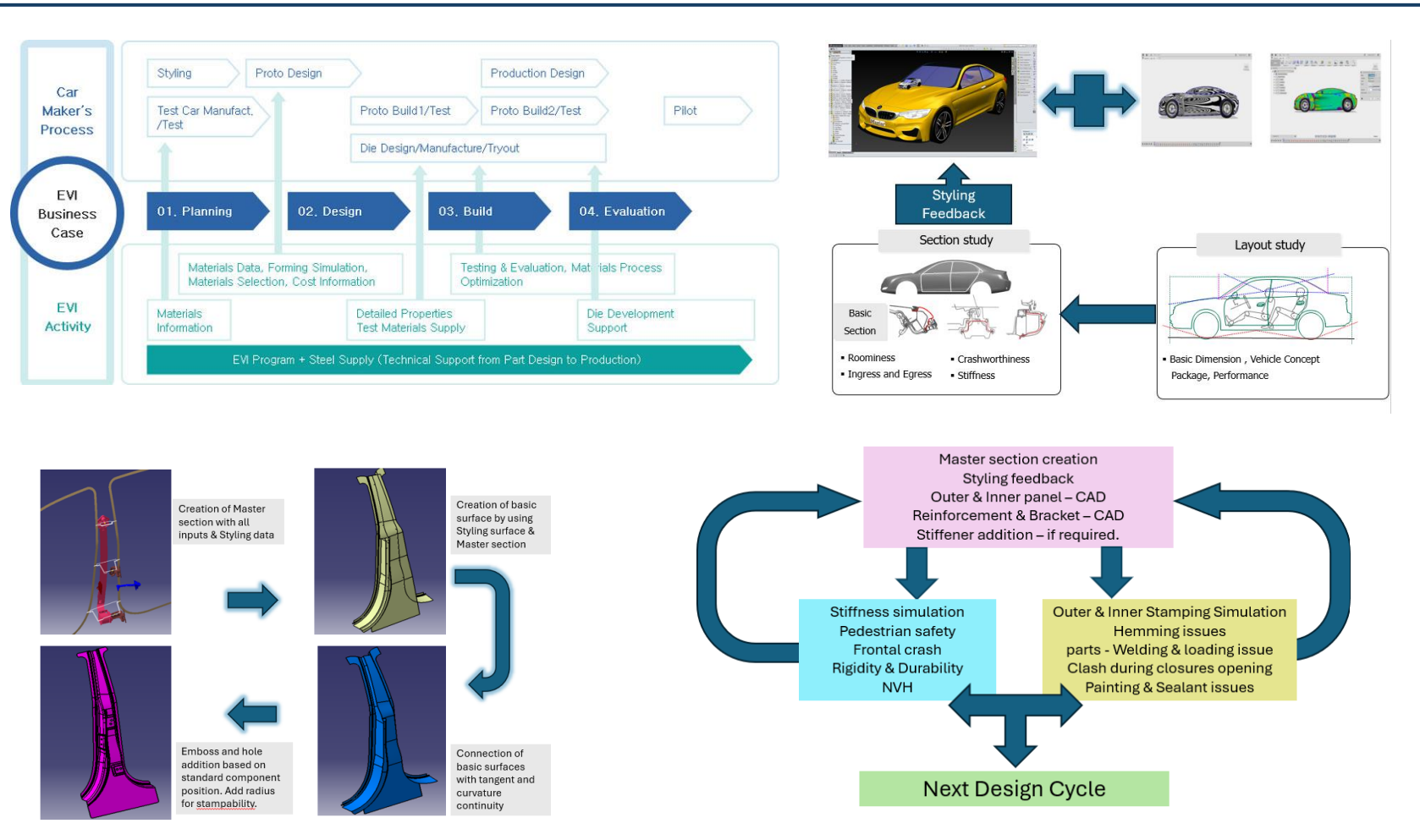
02

Day number

05

Topic name : BIW Vehicle development process

Aim: To learn the BIW vehicle development phase – digital activities



BIW PRODUCT DESIGN COURSE

Week number

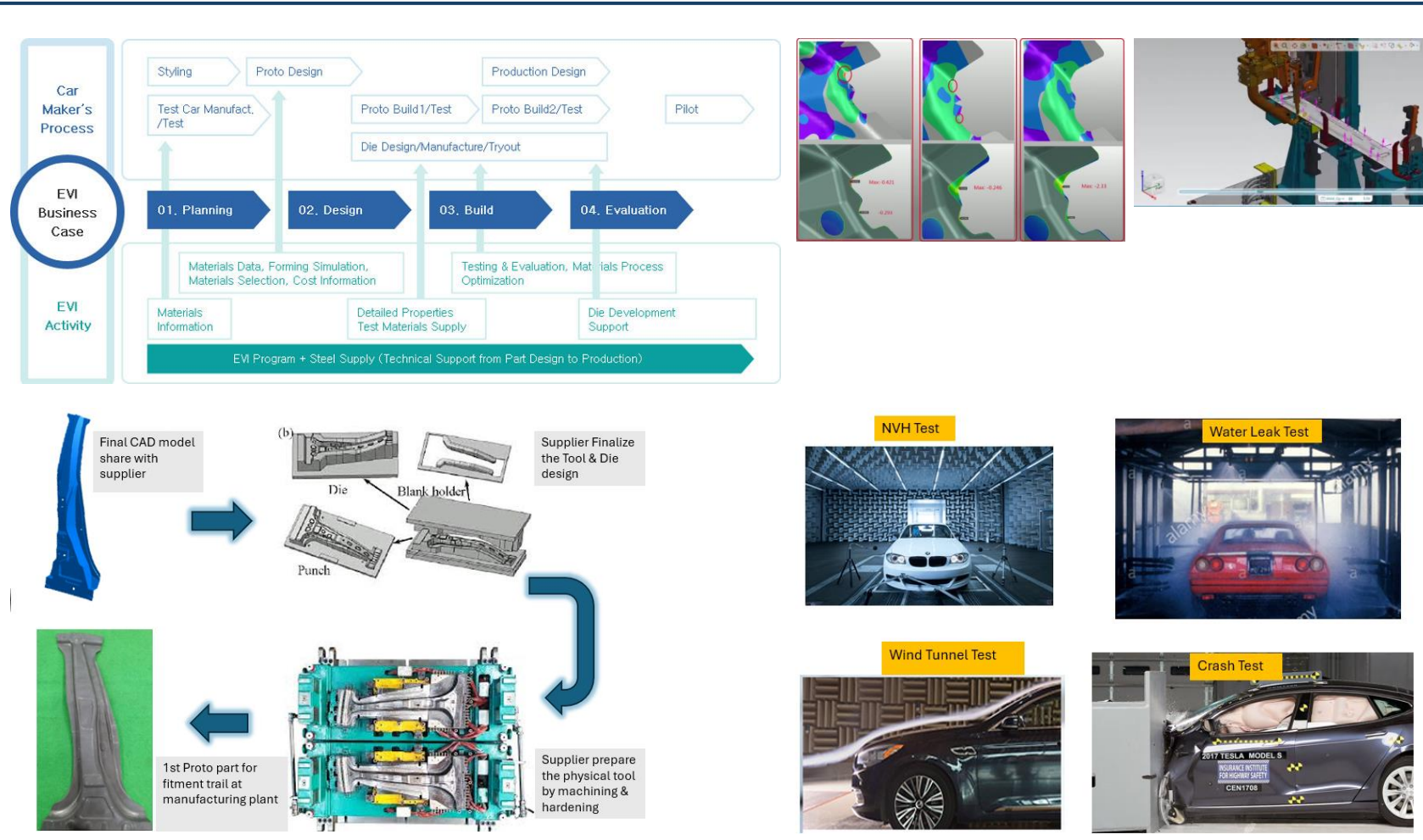
02

Day number

06

Topic name : BIW Vehicle development process

Aim: To learn the BIW vehicle development phase – physical activities



BIW PRODUCT DESIGN COURSE

Week number

03

Day number

07

Topic name : BIW Crashworthiness

Aim: To learn the BIW crash test and regulation

Frontal Impact – Euro NCAP (2015)

- Deformable Barrier
- 64 km/h
- 40% Overlap



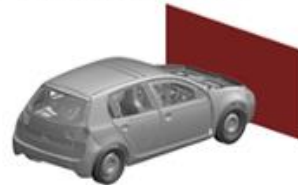
Side Impact Pole – Euro NCAP (2015)

- Rigid Pole
- 32 km/h
- 75°



Frontal Impact – Euro NCAP (2015)

- Rigid Wall
- 50 km/h
- Full Width



Side Impact – Euro NCAP (2015)

- AE Moving Deformable Barrier
- 50 km/h
- 1300 kg



Rear Impact – FMVSS 301 (new)

- Moving Deformable Barrier
- 80 km/h
- 70% Overlap
- 1368 kg



Rollover – FMVSS 216a

- Moving Rigid Plate
- 13 mm/s
- $F = 3 \times 9.8 \text{ m/s}^2 \times m$
($m = m_{\text{veh, unloaded}} < 2722 \text{ kg}$)



BIW PRODUCT DESIGN COURSE

Week number

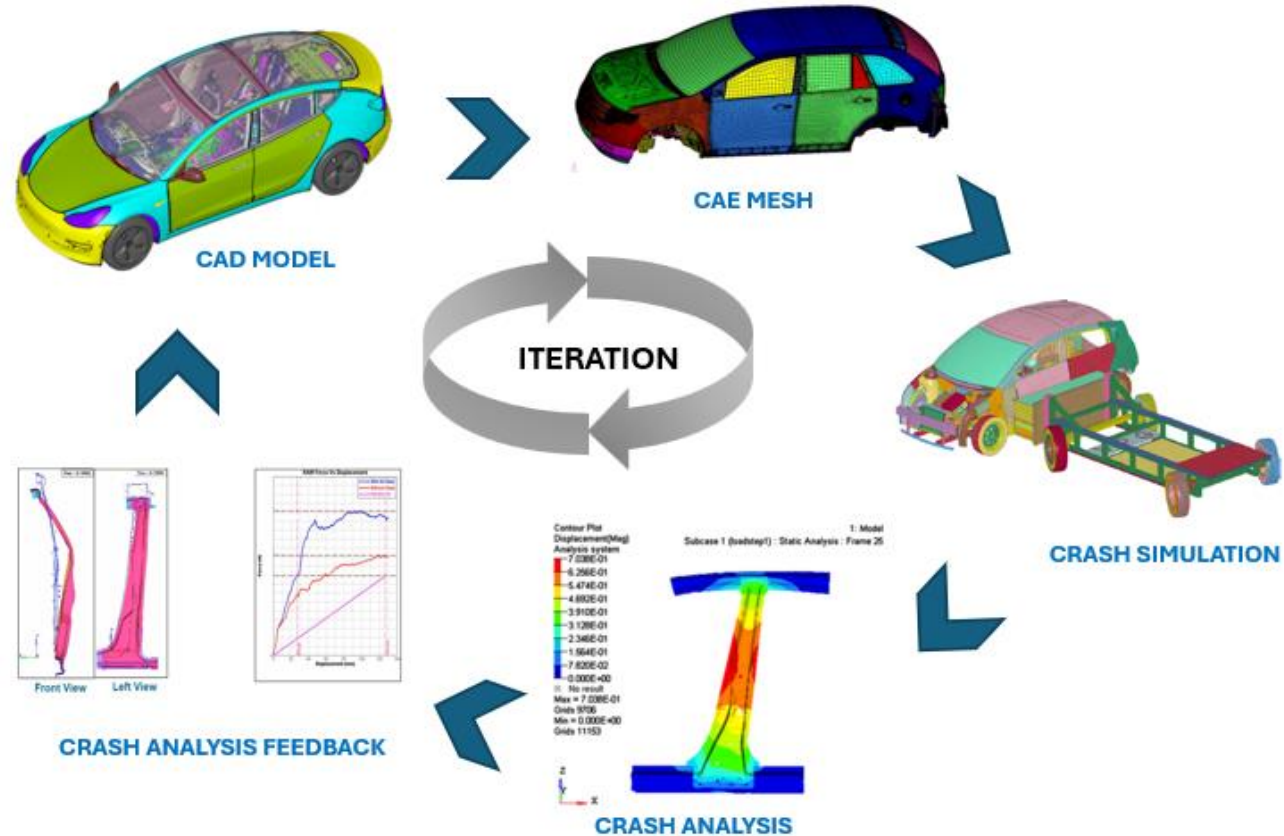
03

Day number

08

Topic name : BIW Crashworthiness

Aim: To learn the how to design BIW to achieve better crash worthiness



BIW PRODUCT DESIGN COURSE

Week number

03

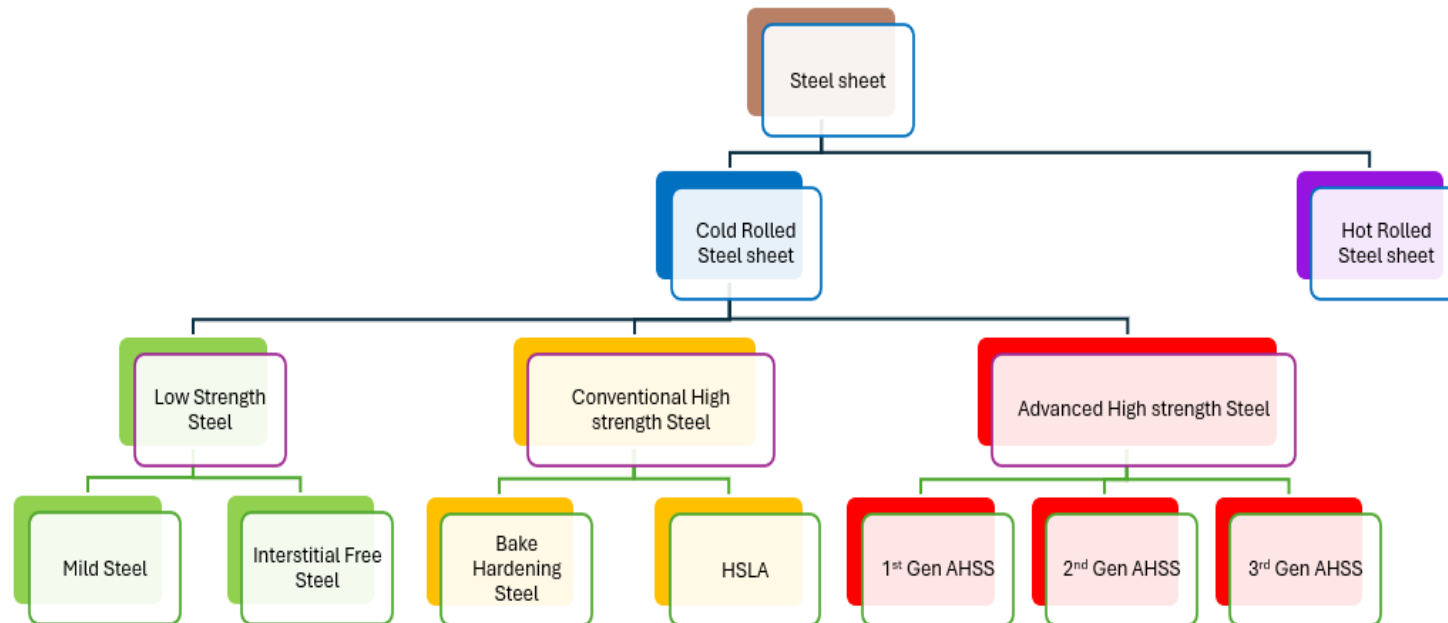
Day number

09

Topic name : BIW Material selection

Aim: To learn about the BIW steel grades & types

Automotive steels can be classified as below



BIW PRODUCT DESIGN COURSE

Week number

04

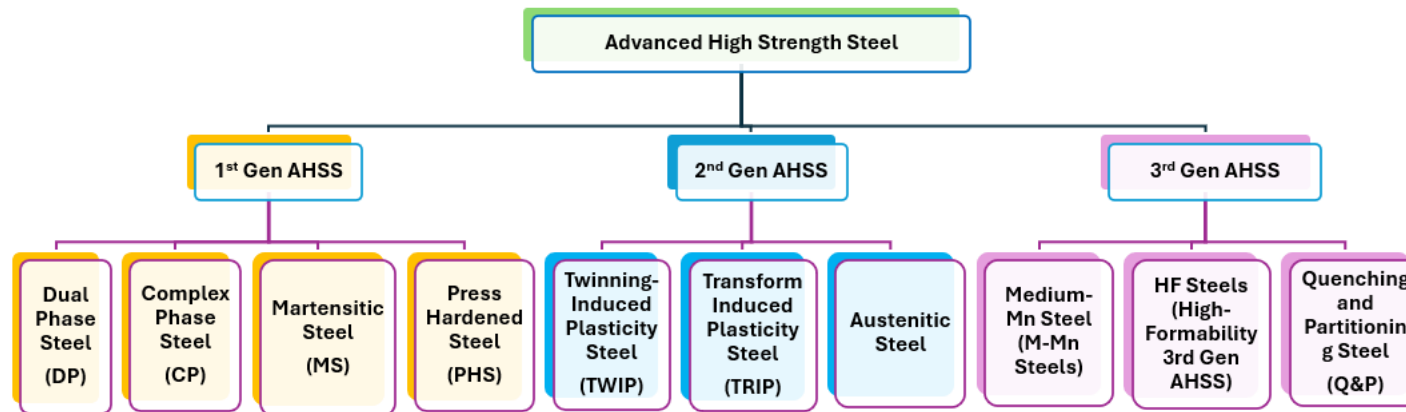
Day number

10

Topic name : BIW Material selection

Aim: To learn about the BIW AHSS steels and AL alloys

Automotive steels can be classified as below



Main Aluminium Alloy Series Used in BIW

Aluminium alloys are grouped by their series, depending on their main alloying elements:

Alloy Series	Major Alloying Element	Typical BIW Use	Key Features
5xxx (Al-Mg)	Magnesium	Outer panels, floor, closures	High formability, medium strength, good corrosion resistance
6xxx (Al-Mg-Si)	Magnesium + Silicon	Structural parts (rails, pillars, crash members)	Heat-treatable, high strength after ageing, good crash performance
7xxx (Al-Zn-Mg)	Zinc + Magnesium	Limited use in high-strength members	Very high strength, lower formability

BIW PRODUCT DESIGN COURSE

Week number

04

Day number

11

Topic name : BIW Manufacturing process

Aim: To learn about the BIW Stamping & welding process

STAMPING PROCESS & STAMPING SEQUENCE



Blanking



Forming



Trimming +
Piercing

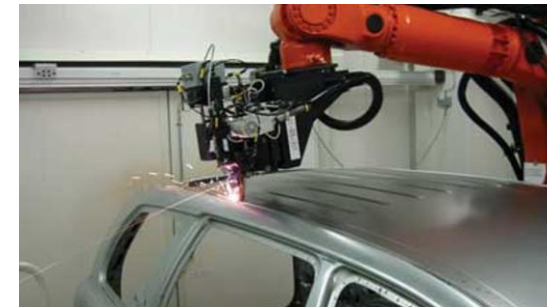
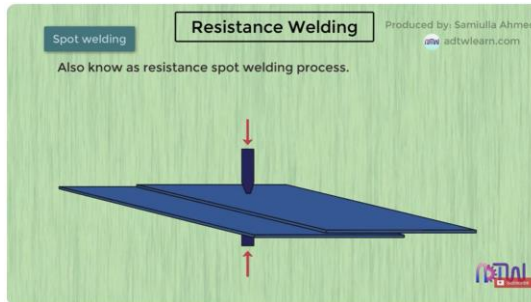


Flanging



Separation

WELDING PROCESS



BIW PRODUCT DESIGN COURSE

Week number

04

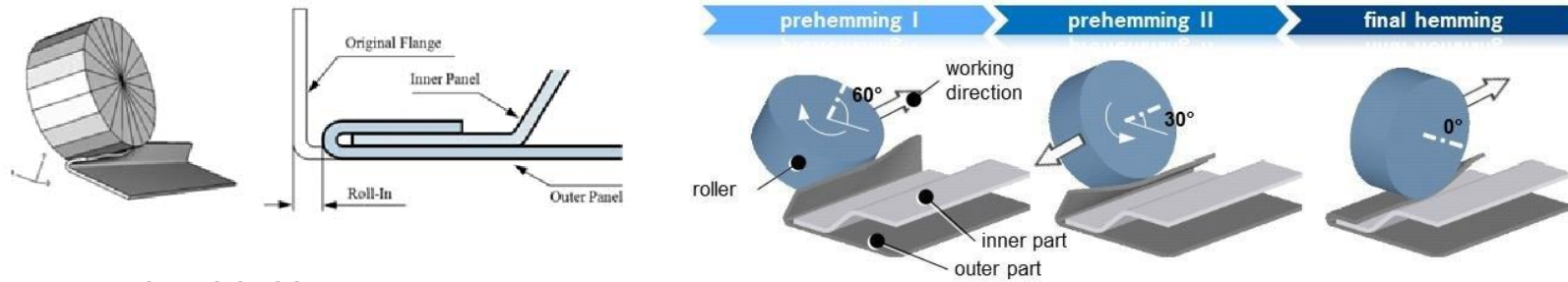
Day number

12

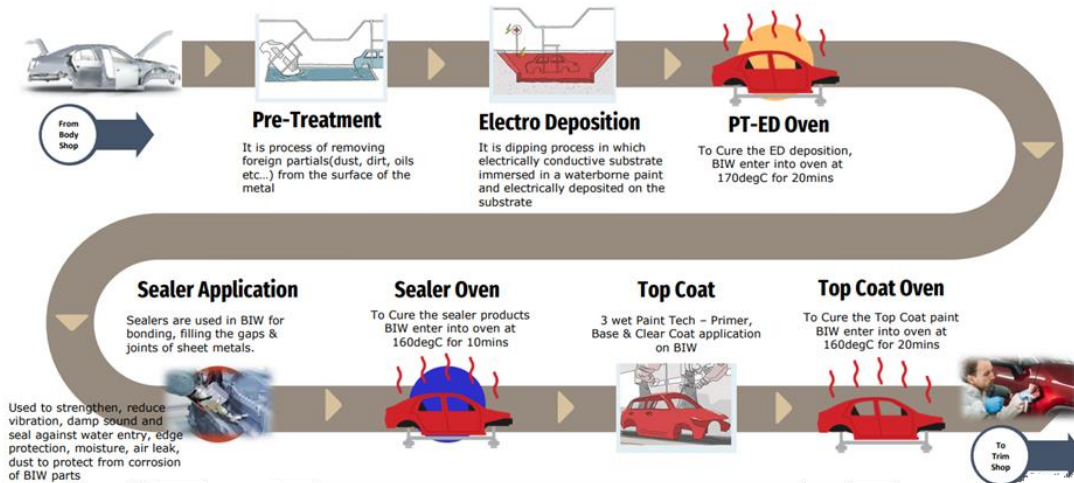
Topic name : BIW Manufacturing process

Aim: To learn about the BIW hemming process, painting process

HEMMING PROCESS



PAINTING PROCESS



BIW PRODUCT DESIGN COURSE

Week number

05

Day number

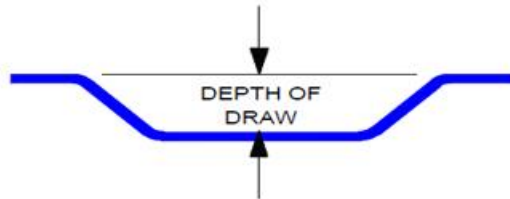
13

Topic name : BIW DFM & DFA

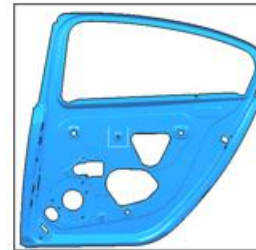
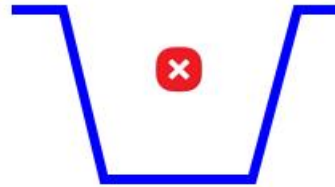
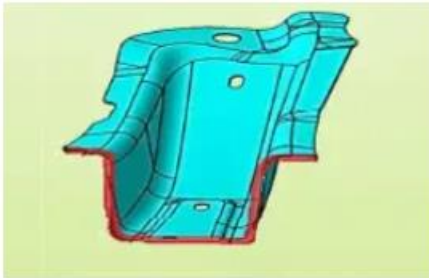
Aim: To learn about the BIW design considerations for stamping

BIW STAMPING PROCESS - DFM

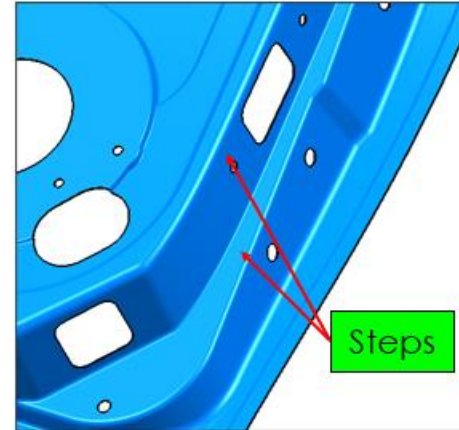
Optimize the Depth of Draw



Avoid deeper draw section for BIW part. It may require more than one additional drawing operation, causing additional die and production costs.



Door require more than one additional drawing operation, causing additional die and production costs.

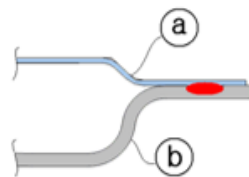


BIW WELDING PROCESS - DFA

Welding Design Consideration

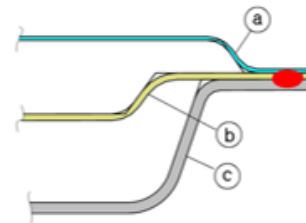
Welding Thickness

- The difference between the thickest and thinnest panel should be triple
- The weld should be thinner than 6.0 mm (2 layer) / 4.5 mm (3 layer)
- Should be avoid more than 3 layer (3 stack) spot welding for better spot weld strength



$$\frac{b}{a} \leq 3$$
$$(a + b) \leq 6.0mm$$

2 layer welding



$$\frac{a + b + c}{a} \leq 4.5$$
$$(a + b + c) \leq 4.6mm$$

3 layer welding

BIW PRODUCT DESIGN COURSE

Week number

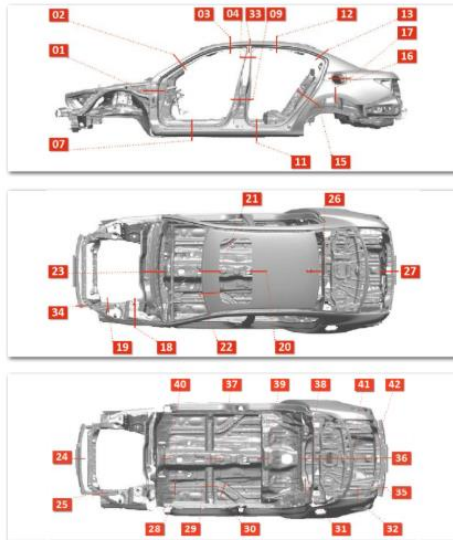
05

Day number

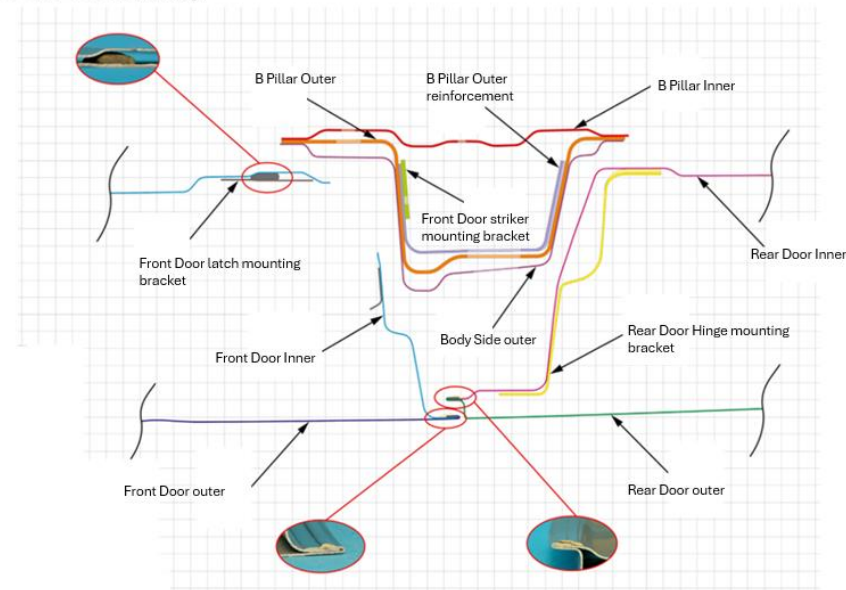
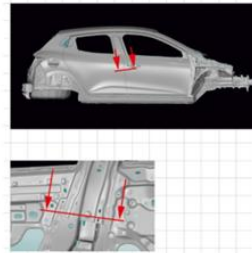
15

Topic name : BIW Master section

Aim: To learn about the BIW master section and requirements



B PILLAR SECTION (FR DOOR LOCK AREA)



BIW PRODUCT DESIGN COURSE

Week number

06

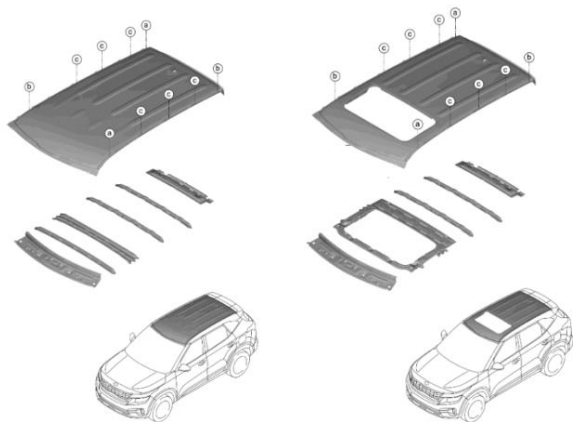
Day number

16

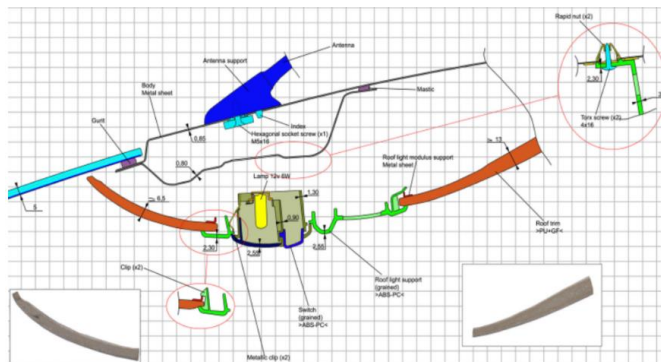
Topic name : ROOF ASSEMBLY PROJECT

Aim: To learn about the BIW Roof assembly introduction & input requirements

Introduction



Benchmarking



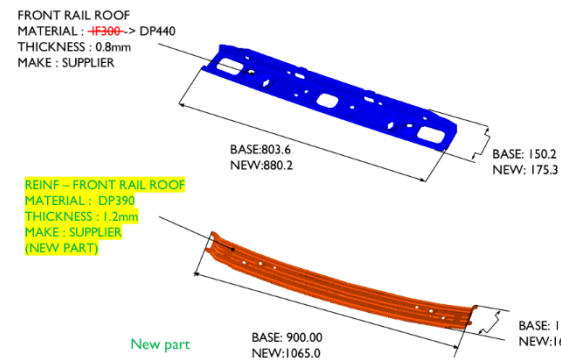
Input Required for Roof design

- Roof surface and Parting line – Styling team
- Vehicle Specification – Project management
- Front visible zone - Vehicle Planning
- Head room conditions - Vehicle Planning
- Head side space conditions - Vehicle Planning
- Overall height - Vehicle Planning
- Sunroof unit conditions - Sunroof unit Design team / Vehicle Architecture
- Trim parts layout – Interior team / Vehicle Architecture

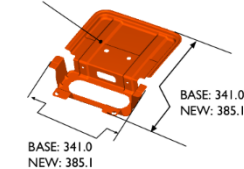
Design Concept sheet (Proposal)

Roof assembly [Normal Roof structure]

FRONT RAIL ROOF
MATERIAL : ~~F360~~ -> DP440
THICKNESS : 0.8mm
MAKE : SUPPLIER



BRKT – MAP LAMP
MATERIAL : ISS13:DD
THICKNESS : ~~1.2mm~~ -> 1.0mm
MAKE : SUPPLIER



17

Aim: To learn about the BIW Front and Rear roof rail master section

BIW PRODUCT DESIGN COURSE

Week number

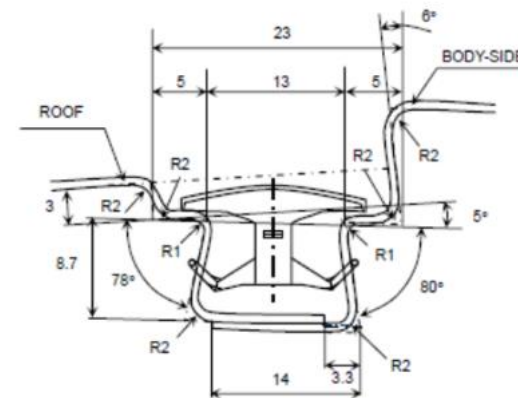
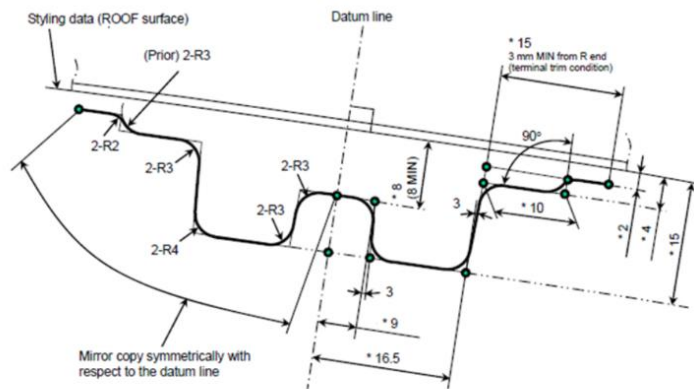
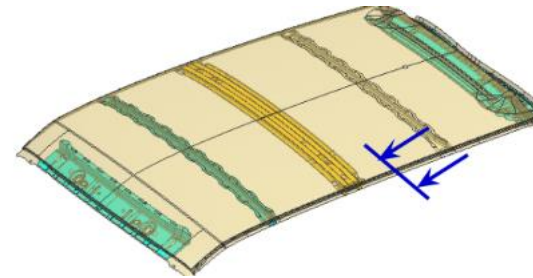
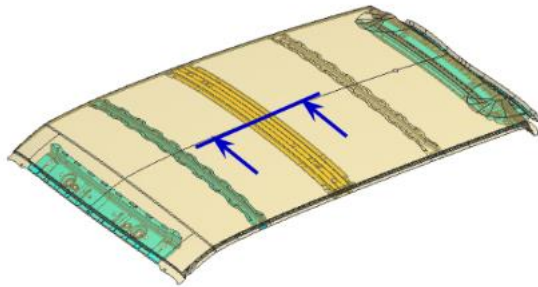
06

Day number

18

Topic name : ROOF ASSEMBLY PROJECT

Aim: To learn about the BIW Roof Bow & Roof drip master section



BIW PRODUCT DESIGN COURSE

Week number

07

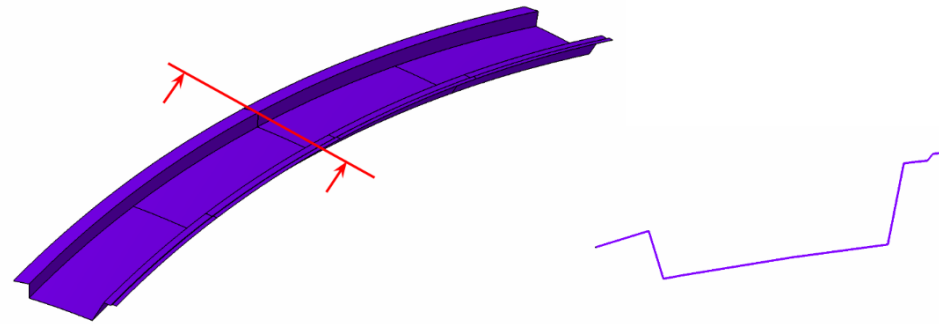
Day number

19

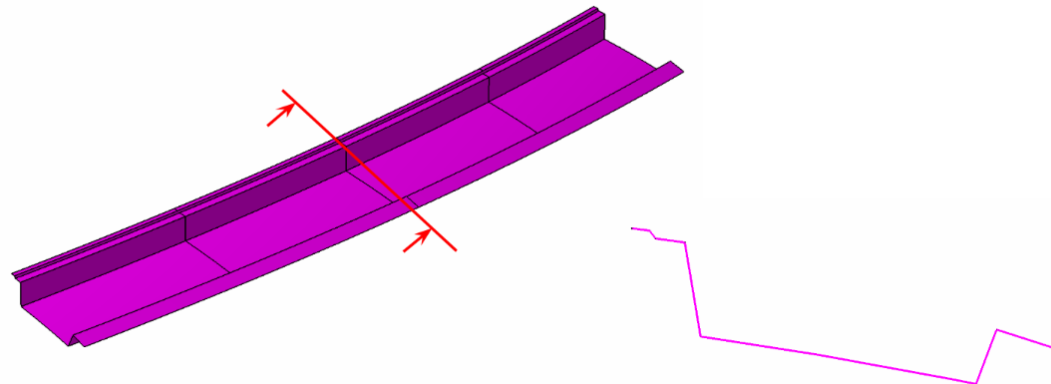
Topic name : ROOF ASSEMBLY PROJECT

Aim: To learn about the BIW CAD creation for Front roof rail & Rear roof rail

FRONT ROOF RAIL - BASIC SHAPE



REAR ROOF RAIL - BASIC SHAPE (For Sedan)



BIW PRODUCT DESIGN COURSE

Week number

07

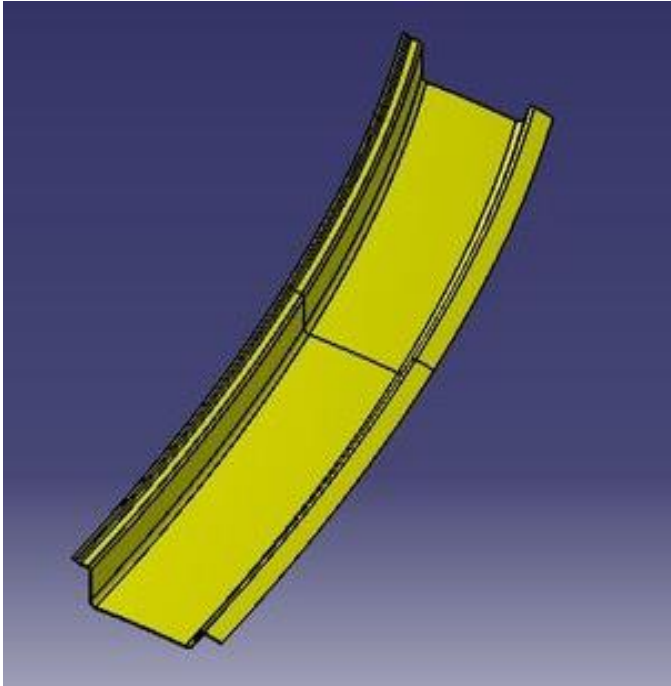
Day number

20

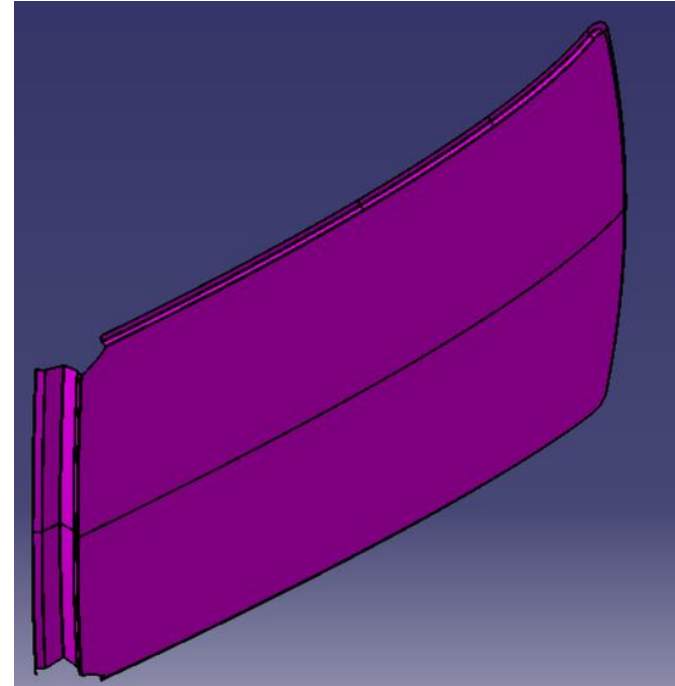
Topic name : ROOF ASSEMBLY PROJECT

Aim: To learn about the BIW Roof panel – Radius & Trimline definition

Front Roof rail radius &
trimline definition



Roof panel side flanges and
connection data creation



BIW PRODUCT DESIGN COURSE

Week number

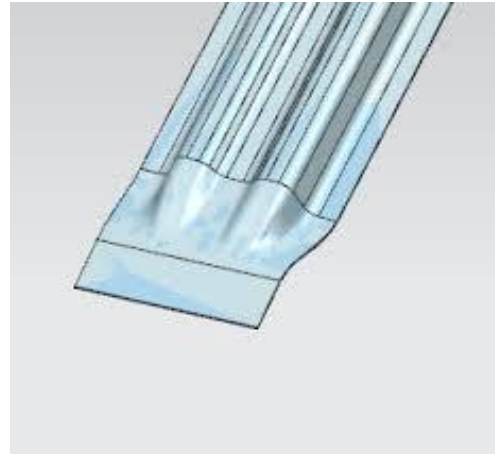
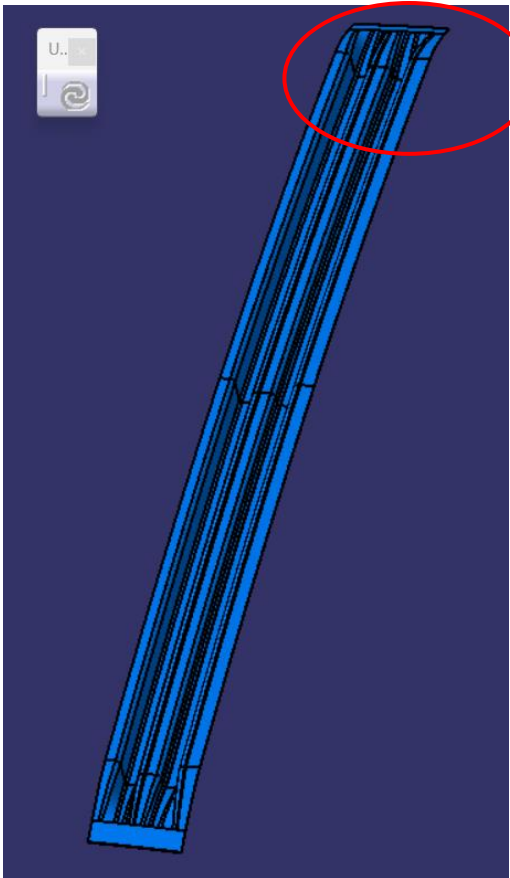
07

Day number

21

Topic name : ROOF ASSEMBLY PROJECT

Aim: To learn about the BIW Roof bow creation



Roof panel side flanges
with Roof bow
connection creation



BIW PRODUCT DESIGN COURSE

Week number

08

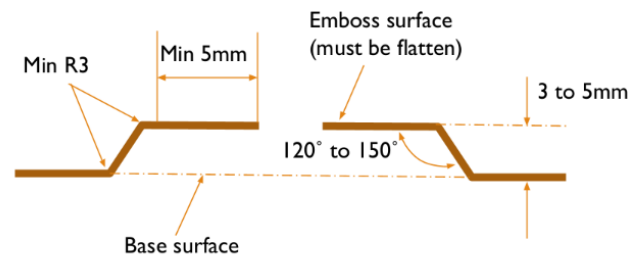
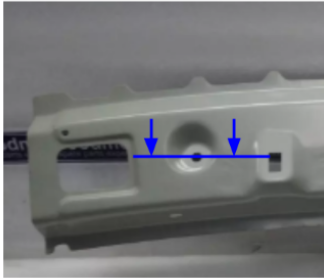
Day number

22

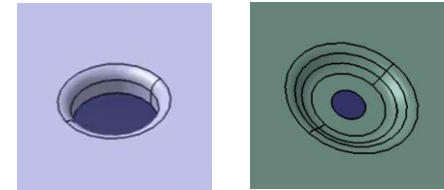
Topic name : ROOF ASSEMBLY PROJECT

Aim: To learn about the BIW Roof panel – Features addition

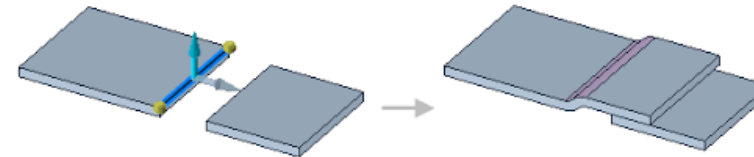
EMBOSS – DESIGN CONSIDERATION



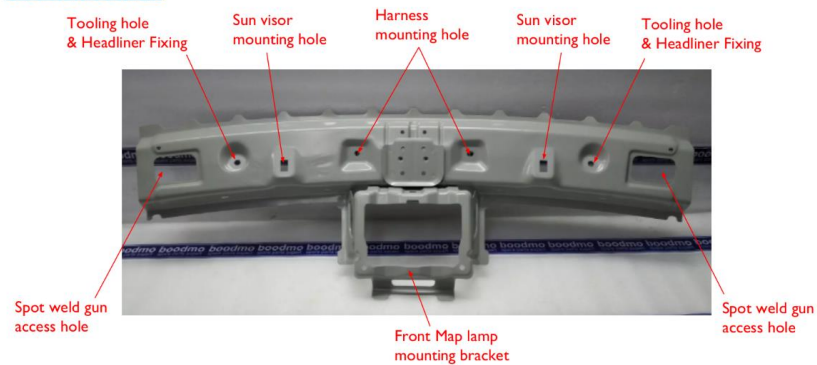
Emboss



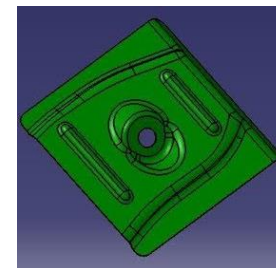
Joggle



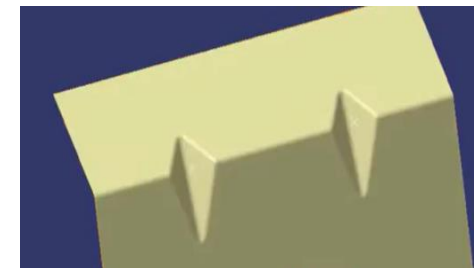
FRONT ROOF RAIL



Diabolo emboss



Stiffener Bead



BIW PRODUCT DESIGN COURSE

Week number

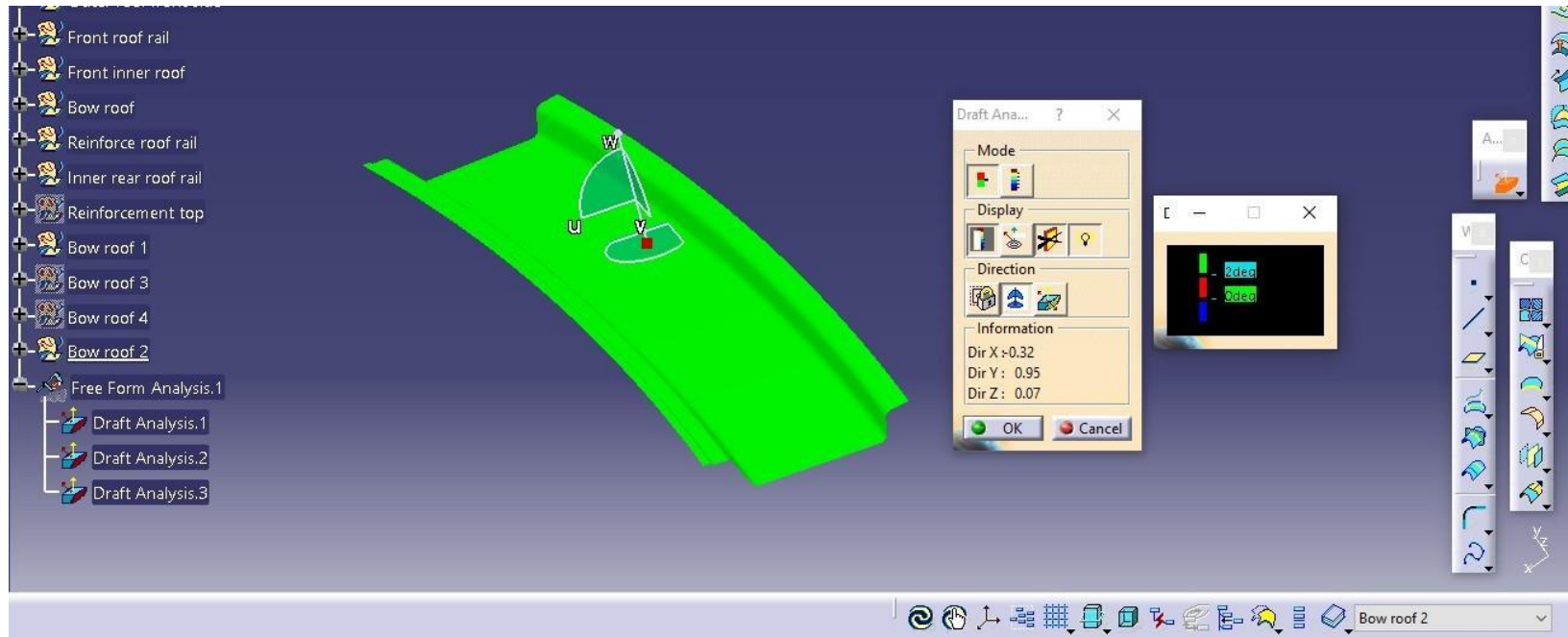
08

Day number

23

Topic name : ROOF ASSEMBLY PROJECT

Aim: To learn about the BIW Front Roof Rail & other parts – Draft analysis



BIW PRODUCT DESIGN COURSE

Week number

08

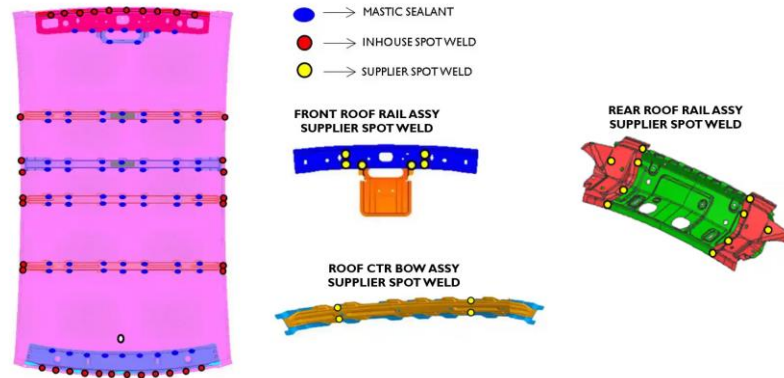
Day number

24

Topic name : ROOF ASSEMBLY PROJECT

Aim: To learn about the BIW Roof - weld creation, BOM Creation & DVP Explanation

Roof Spot weld & Mastic sealant



Load cases	Baseline	Concept 1
Mass	22.4 kg	17.4 Kg
Roof Crush	3.7 (SWR) /62 kN	3.6 (SWR) / 61.4 kN
Frequency- Torsion	50 Hz	50 Hz
Frequency- Bending	37 Hz	36 Hz
Stiffness – Torsion	27.6 kN-m/deg	27.3 kN-m/deg
Stiffness – Bending	6.9 kN/mm	6.8 kN/mm
Dent Resistance (plastic strain)	1.2%	< 1%

LEVEL	PART NUMBER	PART NAME	MATERIAL	THICKNESS	QUANTITY	MASS (KG)	STATUS
1	XXXXXXXXXX	ROOF ASSEMBLY	-			16.34	MAIN ASSEMBLY
2	XXXXXXXXXX	ROOF PANEL	MILD STEEL	0.7	1	9.20	PART
2	XXXXXXXXXX	FRONT ROOF RAIL ASSEMBLY	-				SUB LEVEL ASSEMBLY
3	XXXXXXXXXX	FRONT ROOF RAIL	DP440	1.0	1	0.72	PART
3	XXXXXXXXXX	FRONT ROOF RAIL REINFORCEMENT	DP590	1.6	1	0.85	PART
3	XXXXXXXXXX	MAP LAMP BRACKET	HSLA 340	1.6	1	0.35	PART
2	XXXXXXXXXX	CENTRE ROOF RAIL ASSEMBLY	-				SUB LEVEL ASSEMBLY
3	XXXXXXXXXX	CENTRE ROOF RAIL	DP440	1.2	1	0.82	PART
3	XXXXXXXXXX	CENTRE ROOF RAIL REINFORCEMENT	DP780	1.6	1	1.14	PART
3	XXXXXXXXXX	MAP LAMP BRACKET	HSLA 340	1.6	1	0.35	PART
2	XXXXXXXXXX	REAR ROOF RAIL ASSEMBLY	-				SUB LEVEL ASSEMBLY
3	XXXXXXXXXX	REAR ROOF RAIL	MILD STEEL	1.0	1	0.82	PART
3	XXXXXXXXXX	REAR ROOF RAIL REINFORCEMENT	DP390	1.4	1	1.20	PART
3	XXXXXXXXXX	BACK DOOR HINGE BRACKET LH ASSEMBLY	-				PART LEVEL ASSEMBLY
4	XXXXXXXXXX	BACK DOOR HINGE BRACKET LH	HSLA 340	1.6	1	0.24	PART
4	XXXXXXXXXX	WELD NUT	CI		1	0.01	PART
3	XXXXXXXXXX	BACK DOOR HINGE BRACKET RH ASSEMBLY	-				PART LEVEL ASSEMBLY
4	XXXXXXXXXX	BACK DOOR HINGE BRACKET RH	HSLA 340	1.6	1	0.24	PART
4	XXXXXXXXXX	WELD NUT	CI		1	0.01	PART
2	XXXXXXXXXX	ROOF BOW 1	MILD STEEL	1.0	1	0.37	PART
2	XXXXXXXXXX	ROOF BOW 2	MILD STEEL	1.0	1	0.37	PART

BIW PRODUCT DESIGN COURSE

Week number

09

Day number

25

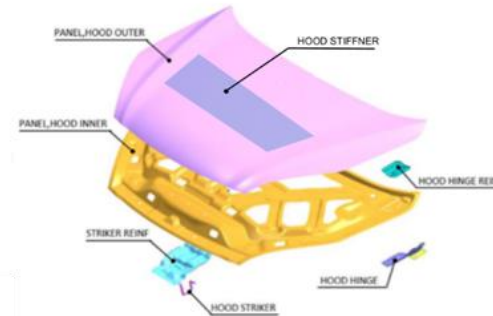
Topic name : HOOD ASSEMBLY PROJECT

Aim: To learn about the BIW Hood assembly introduction & input requirements

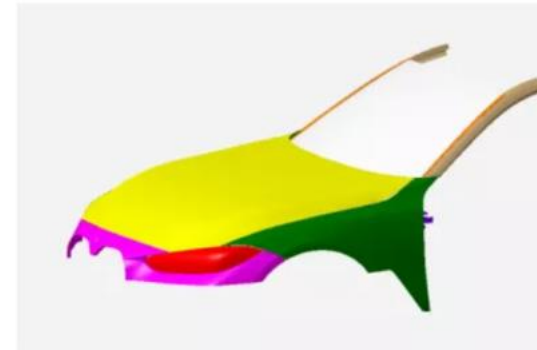
Introduction

HOOD BIW PARTS

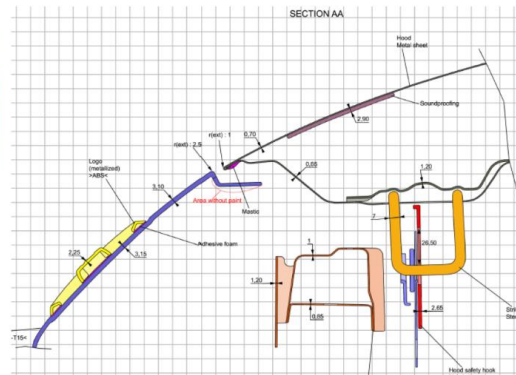
- ☐ Hood Outer Panel
- ☐ Hood Inner Panel
- ☐ Hood Stiffener
- ☐ Hood Hinge Reinforcement
- ☐ Hood Lock Striker Mounting Bracket



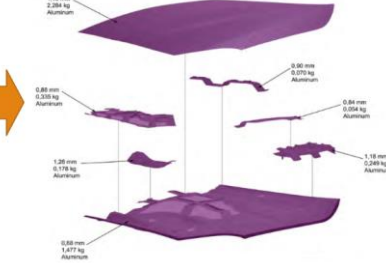
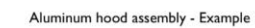
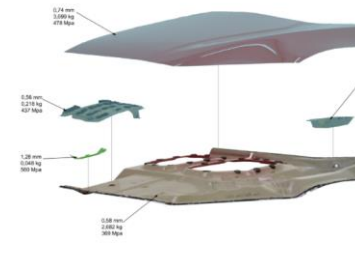
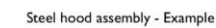
Styling data



Benchmarking



Design Concept Sheet



BIW PRODUCT DESIGN COURSE

Week number

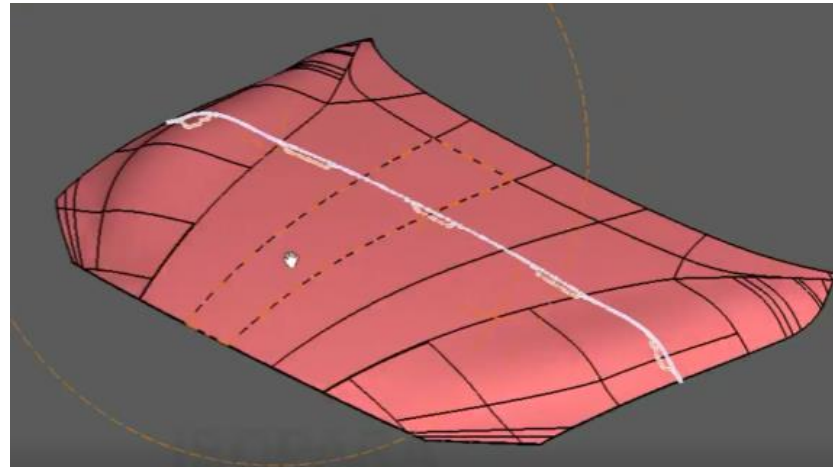
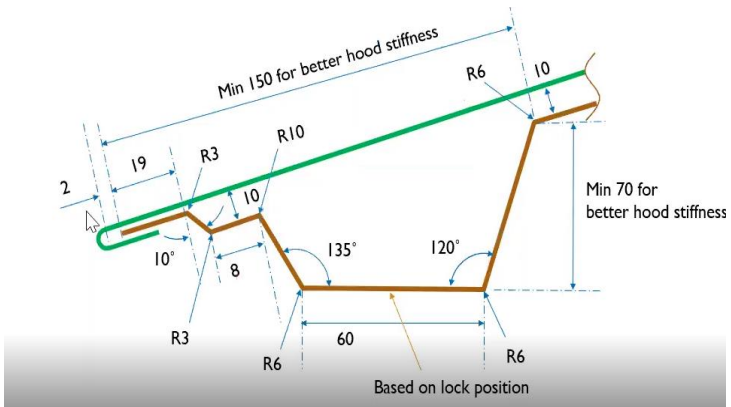
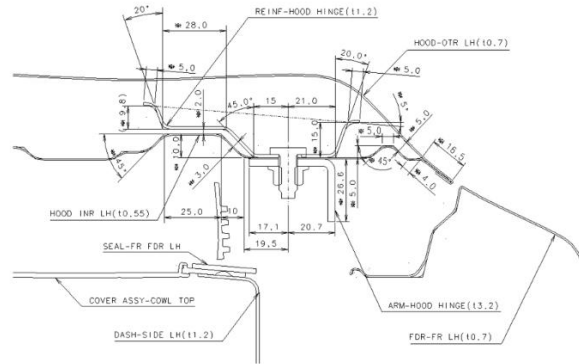
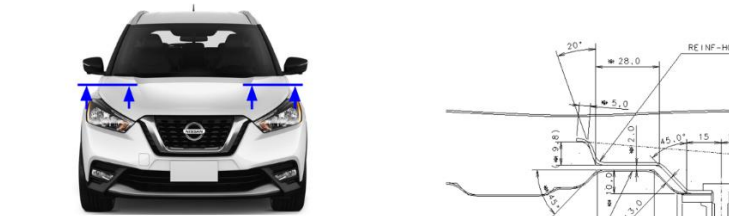
09

Day number

26

Topic name : HOOD ASSEMBLY PROJECT

Aim: To learn about the BIW Hood master section



BIW PRODUCT DESIGN COURSE

Week number

09

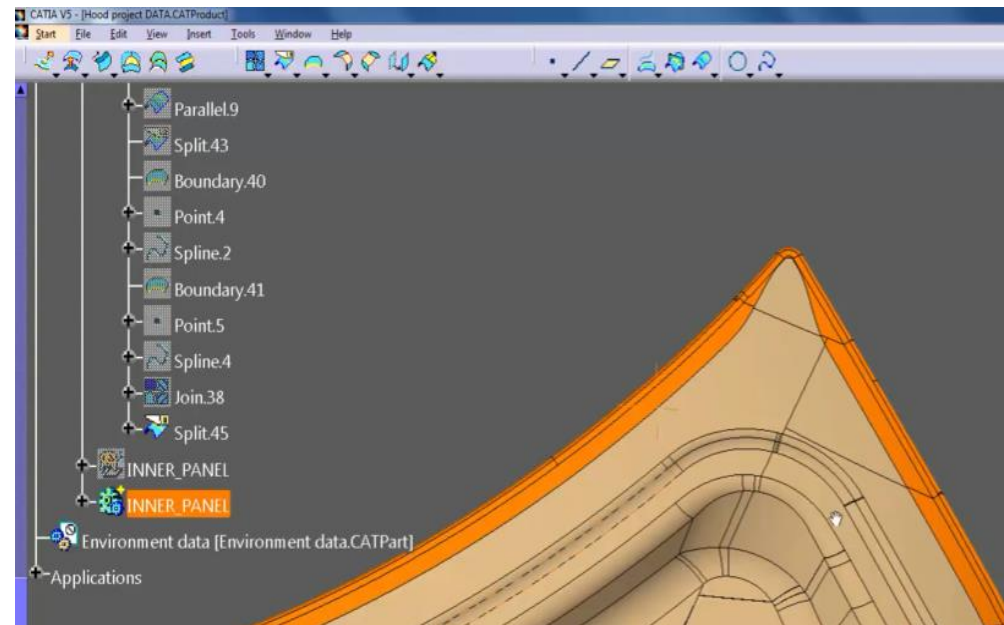
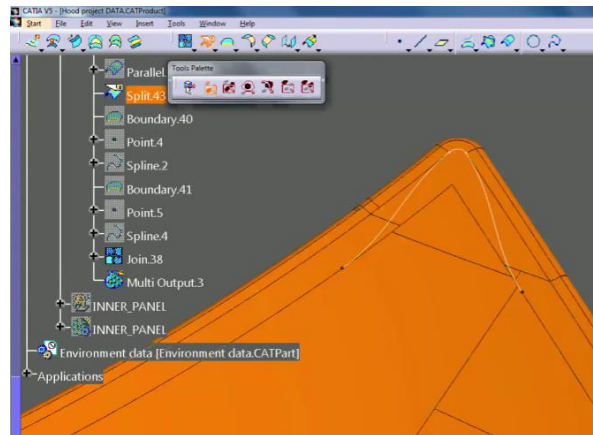
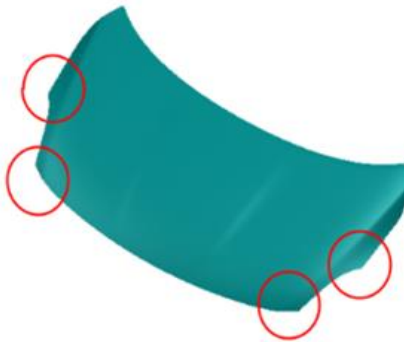
Day number

27

Topic name : HOOD ASSEMBLY PROJECT

Aim: To learn about the BIW Hood outer panel hemming creation

Hood Hemming Shape – Front Side



BIW PRODUCT DESIGN COURSE

Week number

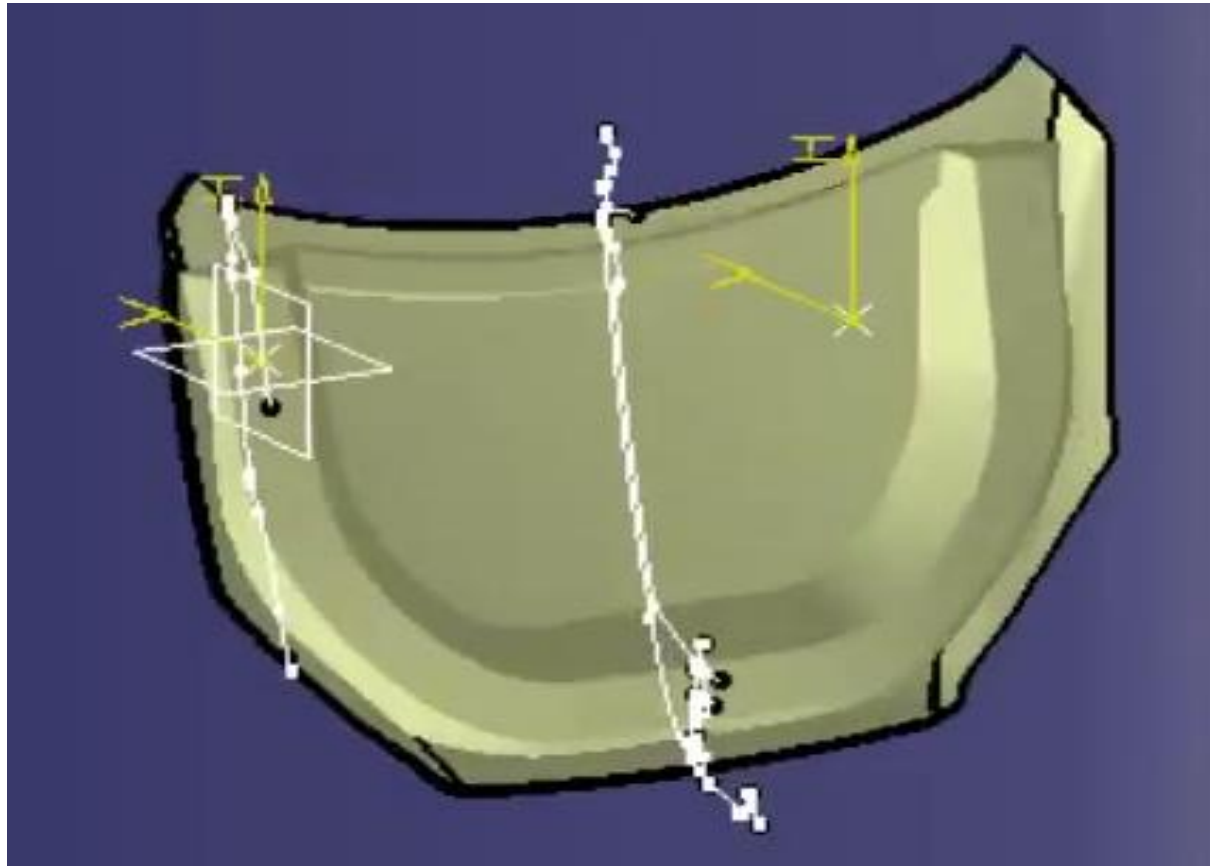
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Day number

28

Topic name : HOOD ASSEMBLY PROJECT

Aim: To learn about the BIW Hood inner panel CAD creation



BIW PRODUCT DESIGN COURSE

Week number

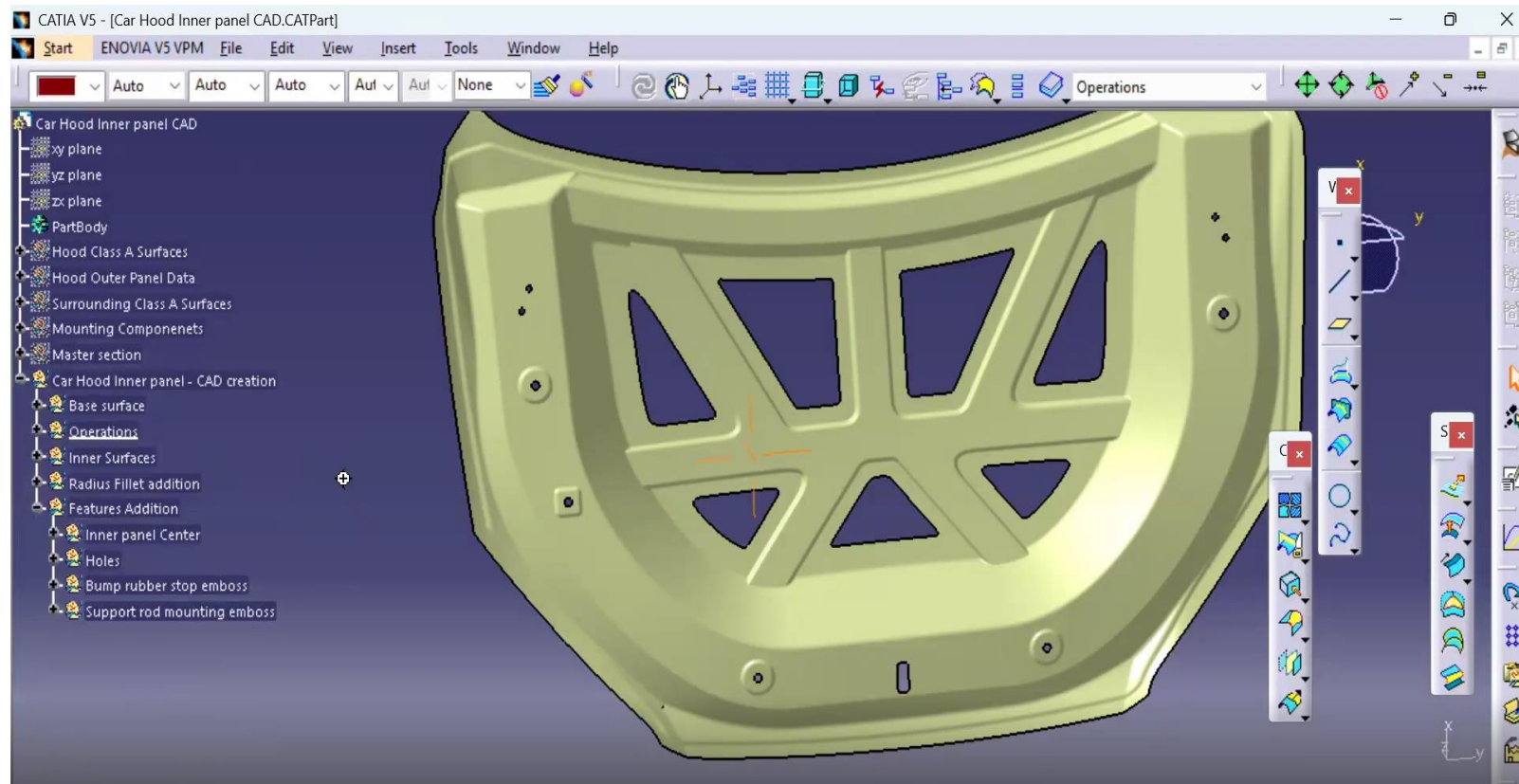
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Day number

29

Topic name : HOOD ASSEMBLY PROJECT

Aim: To learn about the BIW Hood inner panel CAD creation



BIW PRODUCT DESIGN COURSE

Week number

10

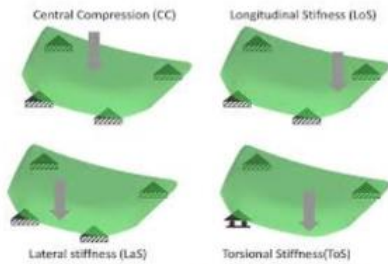
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30

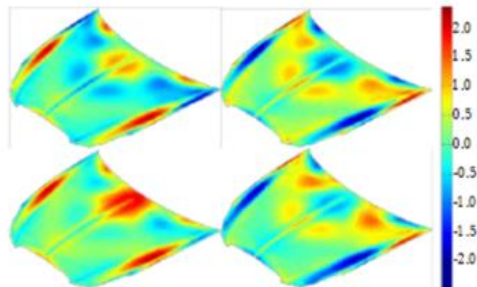
Topic name : HOOD ASSEMBLY PROJECT

Aim: To learn about the BIW Hood assembly – DVP explanation

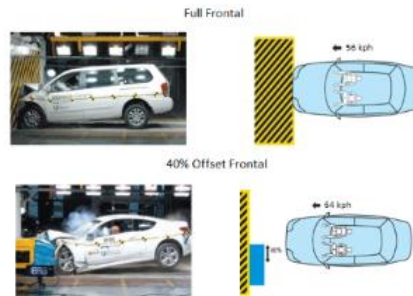
Hood Stiffness test



Hood Stamping Simulation



Hood Crash test



Hood Pedestrian & Child Safety

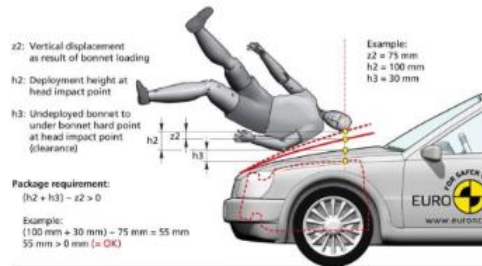
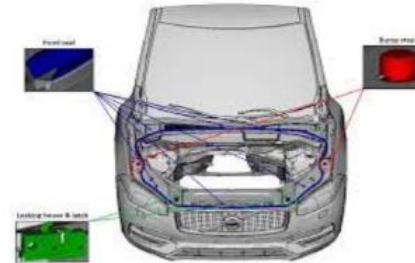


Figure 4: Bonnet deflection total clearance requirement

Hood Slam test



Hood Hinge Durability Study

